S181M116 Derivatives and Alternative Investments: Lecture 2

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This lecture studies two connected questions. First, how can a firm use futures contracts to hedge an existing or anticipated exposure? Second, how are forward and futures prices linked to spot prices, interest rates, income, storage costs, and convenience yields? The main ideas are short and long hedges, basis risk, cross hedging, the minimum-variance hedge ratio, stock-index beta hedging, stack-and-roll hedging, no-arbitrage forward pricing, and commodity cost of carry.

Reading. Hull, Chapters 3 and 5; selected commodity examples from Chapter 35. The central skill is to translate an economic exposure into a futures position, then check whether the hedge is exact, approximate, or exposed to basis risk.

Roadmap

Lecture 1 introduced futures and forwards as contracts. Lecture 2 uses them for two purposes:

1. **Hedging.** Choose a futures position that offsets a price risk.
2. **Pricing.** Use no-arbitrage to connect spot prices, forward prices, interest rates, income, and storage benefits.

The two purposes are linked. A hedge works well when futures and spot prices move together. A pricing formula tells us why they should move together and when they may diverge.

Interpretation. A hedge does not make a firm "win" in all states. It trades upside for downside protection. If the unhedged position would have gained from a favorable price move, the hedge will usually give up much of that gain.

Basic Hedge Mechanics

The Exposure Comes First

Before choosing a futures position, identify the firm's exposure:

- Will the firm *sell* the asset in the future?
- Will the firm *buy* the asset in the future?

- Is the exposure to the same asset as the futures contract?
- What is the quantity, timing, and currency of the exposure?

The hedge should generate gains when the underlying business exposure generates losses.

Definition 1 (Short hedge)
A short hedge uses a short futures position. It is natural when the hedger owns, produces, or expects to receive an asset and will sell it later. The hedge gains when the futures price falls.

Definition 2 (Long hedge)
A long hedge uses a long futures position. It is natural when the hedger expects to buy an asset later. The hedge gains when the futures price rises.

Short Hedge

A short hedge is used by a seller who is worried about a price decline. Examples include:

- an oil producer that will sell crude oil in three months;
- a farmer that will sell grain after harvest;
- a U.S. exporter that will receive euros and later convert them into dollars;
- a portfolio manager who wants to reduce equity-market exposure.

Example 1 (Short hedge for an oil producer)
An oil producer will sell 500,000 barrels of crude oil in three months. The three-month futures price is USD 78 per barrel. Each futures contract is for 1,000 barrels, so the firm shorts

$$N = \frac{500{,}000}{1{,}000} = 500$$

contracts.

Suppose that when the oil is sold, the spot and futures prices are both close to USD 72. The firm sells physical oil for

$$500{,}000 \times 72 = \$36{,}000{,}000.$$

The short futures gain is

$$500{,}000(78 - 72) = \$3{,}000{,}000.$$

Total revenue is

$$36{,}000{,}000 + 3{,}000{,}000 = \$39{,}000{,}000,$$

or USD 78 per barrel.

If instead the final price is USD 84, the firm sells oil for USD 42 million and loses USD 3 million on futures. Total revenue is again USD 39 million. The hedge has locked in approximately the futures price.

Long Hedge

A long hedge is used by a buyer who is worried about a price increase. Examples include:

- an airline that will buy jet fuel;
- a food producer that will buy wheat;
- a manufacturer that will buy copper;
- an importer that will need foreign currency.

Example 2 (Long hedge for a copper buyer)
A manufacturer will need 250,000 pounds of copper in four months. The futures price is USD 3.70 per pound and each futures contract covers 25,000 pounds. The firm goes long

$$N = \frac{250{,}000}{25{,}000} = 10$$

contracts.

If copper later costs USD 4.10 per pound, the physical purchase costs

$$250{,}000 \times 4.10 = \$1{,}025{,}000.$$

The long futures gain is

$$250{,}000(4.10 - 3.70) = \$100{,}000.$$

The net cost is therefore

$$1{,}025{,}000 - 100{,}000 = \$925{,}000,$$

or USD 3.70 per pound.

If the final price is USD 3.40, the firm pays less in the physical market but loses on futures. The hedge again produces an approximate net purchase price of USD 3.70 per pound.

Hedge-and-Forget Versus Dynamic Hedging

In this lecture, the hedge is usually established at the start and closed at the end. This is a *hedge-and-forget* strategy. Later in the course, option hedging requires dynamic rebalancing because option exposures change with the underlying price, volatility, and time.

> **Rule of thumb.** Linear exposures can often be hedged with a fixed futures position. Nonlinear exposures usually require dynamic hedging or option contracts.

Why Firms Hedge

Risk Reduction

Many nonfinancial firms are not in the business of forecasting oil prices, exchange rates, interest rates, or equity indices. Hedging lets managers focus on operating decisions rather than market timing.

Example 3 (Budget certainty)
An airline sells tickets months before it buys the fuel needed for those flights. If fuel prices rise sharply, the revenue from ticket sales may be fixed while costs increase. A long futures hedge can stabilize the fuel cost per flight and make route planning more reliable.

Hedging and Shareholders

A common argument against corporate hedging is that diversified shareholders can hedge or diversify risk themselves. This argument is incomplete for several reasons:

- managers usually know more about the firm's exposures than outside shareholders;
- firms may hedge at lower transaction cost than individuals;
- reducing cash-flow volatility can reduce financial distress costs;
- stable cash flows can protect investment plans from funding shocks.

The strongest case against hedging is when the risk is already naturally passed through to customers or offset elsewhere in the firm's economics.

Hedging and Competitors

The correct hedge depends on the whole business model. If all firms in an industry are exposed to the same input cost and output prices adjust with that input cost, hedging can sometimes increase rather than decrease profit-margin volatility relative to competitors.

Example 4 (Input cost pass-through)
Suppose jewelry producers buy gold and sell jewelry. If gold prices rise, the wholesale price of jewelry may also rise. A producer that does not hedge may maintain a roughly stable margin. A producer that hedges gold purchases may gain on the hedge when gold rises, but may lose relative to the natural pass-through when gold falls. The hedge must be evaluated against total firm profit, not only against the input price.

Governance issue. A good hedge can look bad ex post. If a short hedge loses money because the asset price rose, the underlying business exposure should have gained. Risk reports should show the combined physical-plus-futures result, not the futures account in isolation.

Basis Risk

The Basis

The *basis* is the difference between the spot price of the asset being hedged and the futures price of the contract used for hedging:

$$b_t = S_t - F_t.$$

Some markets define the basis as futures minus spot. In this course note, unless otherwise stated, we use

$$\text{basis} = \text{spot} - \text{futures}.$$

Basis risk is the risk that the basis at the time the hedge is closed is not the basis expected when the hedge is initiated.

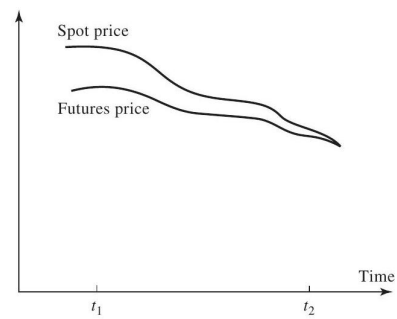

Figure 1: Spot and futures prices usually move together, but the gap between them can change before convergence. That changing gap is the basis risk.

Effective Price in a Short Hedge

Suppose a firm will sell the asset at time t_2. It enters a short futures position at time t_1 at futures price F_1 and closes it at F_2. The spot price at t_2 is S_2. The effective sale price is

$$S_2 + (F_1 - F_2).$$

Using $b_2 = S_2 - F_2$, this becomes

$$S_2 + F_1 - F_2 = F_1 + b_2.$$

The initial futures price F_1 is known at hedge initiation. The final basis b_2 is not known. Therefore the hedge locks in F_1 plus an uncertain basis.

Example 5 (Short hedge with basis risk)
A grain producer shorts futures at $F_1 = 6.20$ dollars per bushel. At harvest, the local cash price is $S_2 = 5.85$ and the futures price is $F_2 = 5.95$. The final basis is

$$b_2 = 5.85 - 5.95 = -0.10.$$

The effective sale price is

$$F_1 + b_2 = 6.20 - 0.10 = 6.10.$$

Equivalently, the physical sale gives 5.85 and the short futures gain is $6.20 - 5.95 = 0.25$, so the effective price is 6.10.

Effective Price in a Long Hedge

Suppose a firm will buy the asset at time t_2. It enters a long futures position at F_1 and closes it at F_2. The effective purchase price is

$$S_2 - (F_2 - F_1) = F_1 + b_2.$$

Again, the uncertainty is the final basis.

Example 6 (Long hedge with basis risk)
A manufacturer goes long copper futures at $F_1 = 3.70$. When the hedge is closed, the physical copper price is $S_2 = 3.92$ and the futures price is $F_2 = 3.88$. The final basis is

$$b_2 = 3.92 - 3.88 = 0.04.$$

The effective purchase price is

$$F_1 + b_2 = 3.70 + 0.04 = 3.74.$$

The physical purchase is more expensive than 3.70 because the final basis is positive.

Basis Strengthening and Weakening

The basis strengthens when it increases. It weakens when it decreases.

Basis movement	Short hedge	Long hedge
Basis strengthens	improves effective sale price	worsens effective purchase price
Basis weakens	worsens effective sale price	improves effective purchase price

Sources of Basis Risk

Basis risk arises when:

- the asset being hedged is not exactly the futures underlying;
- the hedge date does not match the futures delivery date;
- the delivery location or grade differs from the firm's physical exposure;
- local supply-demand conditions move cash prices differently from exchange futures prices;
- the hedge has to be rolled from one contract into another.

> **Contract choice.** A common rule is to choose a delivery month as close as possible to, but later than, the hedge expiration date. The rule must be balanced against liquidity and delivery-risk concerns.

Cross Hedging

When Cross Hedging Is Needed

A cross hedge uses a futures contract on one asset to hedge exposure to another asset. It is used when no liquid futures contract exists for the exact exposure.

Example 7 (Jet fuel and heating oil)
An airline wants to hedge jet fuel. If the jet-fuel futures market is illiquid, the airline might use heating-oil futures because the two prices are strongly related. The hedge is imperfect because the jet-fuel spread over heating oil can change.

If S_2 is the price of the asset being hedged and S_2^* is the spot price of the asset underlying the futures contract, the final basis can be decomposed as

$$S_2 - F_2 = (S_2 - S_2^*) + (S_2^* - F_2).$$

The first term is the difference between the two assets. The second is the basis for the futures underlying itself.

Minimum-Variance Hedge Ratio

Let ΔS be the change in the spot price of the asset being hedged over the hedge horizon. Let ΔF be the change in the futures price. If the hedger uses hedge ratio h, a short hedge has change

$$\Delta V = \Delta S - h\Delta F.$$

The variance of the hedged position is

$$\mathrm{Var}(\Delta V) = \sigma_S^2 + h^2\sigma_F^2 - 2h\,\mathrm{Cov}(\Delta S, \Delta F).$$

Differentiate with respect to h and set the derivative equal to zero:

$$2h\sigma_F^2 - 2\,\mathrm{Cov}(\Delta S, \Delta F) = 0.$$

Thus the minimum-variance hedge ratio is

$$h^* = \frac{\mathrm{Cov}(\Delta S, \Delta F)}{\mathrm{Var}(\Delta F)} = \rho\frac{\sigma_S}{\sigma_F},$$

where ρ is the correlation between ΔS and ΔF.

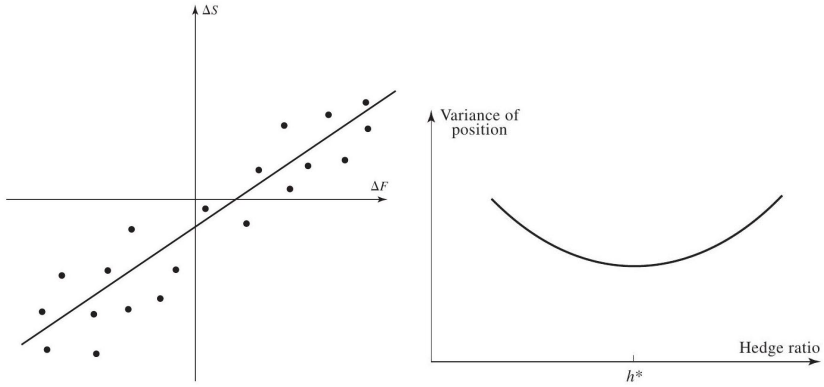

Figure 2: The minimum-variance hedge ratio can be estimated as a regression slope. The resulting hedge variance is minimized at h^*.

Regression interpretation. The hedge ratio h^* is the slope in a regression of spot-price changes on futures-price changes:

$$\Delta S = a + h^* \Delta F + \varepsilon.$$

The hedge effectiveness is the fraction of variance removed by the hedge. In the simple regression setting it is $R^2 = \rho^2$.

Number of Contracts

Let Q_A be the size of the exposure and Q_F the size of one futures contract. The optimal number of contracts is

$$N^* = h^* \frac{Q_A}{Q_F}.$$

If the exposure and futures contract are measured by value rather than physical quantity, use

$$N^* = h^* \frac{V_A}{V_F},$$

where V_A is the value of the exposure and V_F is the value of one futures contract.

Example 8 (Cross hedge ratio)
An airline estimates that monthly jet-fuel price changes have standard deviation 0.18 *dollars per gallon, heating-oil futures price changes have standard deviation* 0.15*, and the correlation is* 0.80*. The minimum-variance hedge ratio is*

$$h^* = 0.80 \frac{0.18}{0.15} = 0.96.$$

The airline should hedge 96% of its physical quantity using heating-oil futures. The hedge ratio is below or above one depending on relative volatilities and correlation; it is not mechanically one.

Example 9 (Optimal number of contracts)
The airline wants to hedge 3 million gallons of jet fuel. Each heating-oil futures contract covers 42,000 gallons. With $h^* = 0.96$,

$$N^* = 0.96 \frac{3{,}000{,}000}{42{,}000} \approx 68.57.$$

The airline would choose 68 or 69 contracts, then consider liquidity, transaction costs, and whether the hedge should be split across maturities.

Percentage-Change Hedge Ratios

Sometimes the regression uses percentage price changes rather than dollar price changes. If $\hat{h}$ is the slope from percentage changes, then the contract count should be based on values:

$$N^* = \hat{h} \frac{V_A}{V_F}.$$

This is useful for equity-index hedging, where beta is naturally defined using returns.

Stock Index Futures and Beta Hedging

Index Futures

A stock index futures contract gives exposure to a diversified stock index. The contract value is

$$V_F = \text{futures price} \times \text{contract multiplier}.$$

For example, if an index futures price is 5,200 and the multiplier is USD 50, one contract has notional value

$$5{,}200 \times 50 = \$260{,}000.$$

Hedging an Equity Portfolio

If a portfolio mirrors the index, the hedge ratio is approximately one. If the portfolio has beta β relative to the index, the number of contracts to short is

$$N^* = \beta \frac{V_A}{V_F}.$$

Example 10 (Equity portfolio hedge)
A portfolio is worth USD 40 million and has beta 1.25 relative to an index. The index futures price is 4,000 and the contract multiplier is USD 250. One contract has notional value

$$V_F = 4{,}000 \times 250 = \$1{,}000{,}000.$$

The number of contracts required to hedge market risk is

$$N^* = 1.25\frac{40{,}000{,}000}{1{,}000{,}000} = 50.$$

The portfolio manager shorts 50 futures contracts.

Changing Portfolio Beta

Futures can also change, rather than eliminate, market exposure. If the current portfolio beta is β and the target beta is β^*, then

$$N^* = (\beta - \beta^*)\frac{V_A}{V_F}.$$

If $N^* > 0$, short futures. If $N^* < 0$, go long futures.

Example 11 (Reducing beta without selling stocks)
A manager has a USD 25 million portfolio with beta 1.40 and wants beta 0.80 for the next month. Each index futures contract has notional value USD 250,000. The number of contracts is

$$N^* = (1.40 - 0.80)\frac{25{,}000{,}000}{250{,}000} = 60.$$

The manager shorts 60 contracts.

Example 12 (Increasing beta)
A manager has a USD 10 million portfolio with beta 0.70 and wants beta 1.10. One futures contract has notional value USD 200,000:

$$N^* = (0.70 - 1.10)\frac{10{,}000{,}000}{200{,}000} = -20.$$

The negative sign means the manager should go long 20 contracts.

Reasons to Use Index Futures

Index futures can:

- temporarily reduce market exposure without selling the underlying stocks;
- equitize cash during portfolio transitions;
- change beta quickly and cheaply;
- separate stock selection from market timing.

Residual risk. A beta hedge removes exposure to broad market moves only approximately. The portfolio can still underperform because of stock selection, sector tilts, changing beta, dividends,

liquidity, and futures basis risk.

Stack-and-Roll Hedging

The Problem

Sometimes the exposure lasts longer than the liquid futures contracts available. The firm then hedges with a nearby contract and rolls the hedge forward as the nearby contract approaches maturity. This is called *stack and roll*.

Example 13 (Rolling a short hedge)
An oil producer wants to hedge sales over the next 18 months, but the most liquid futures contracts are in the first six months. The firm may short nearby contracts, close them as they approach maturity, and open new short positions in later contracts. The hedge is repeatedly rolled forward.

Roll Risk

Rolling creates risk because the contract being closed and the new contract being opened may have changed in relative price. If the term structure moves unfavorably, the firm can suffer large cash-flow losses even when the long-run economics of the hedge are defensible.

Risk-management lesson. A stack-and-roll hedge can transform price risk into liquidity and funding risk. Daily settlement matters because losses on the futures position require cash before the physical exposure generates offsetting cash flows.

Forward Prices for Investment Assets

Investment Assets and Consumption Assets

An *investment asset* is held by many investors primarily for investment purposes. Examples include stocks, bonds, gold, and silver. A *consumption asset* is held primarily for use in production or consumption. Examples include copper, oil, natural gas, wheat, and electricity.

This distinction matters because investment assets can usually be priced by cash-and-carry arbitrage. Consumption assets can have convenience benefits from physical ownership that are not captured by a futures contract.

Assumptions

The standard no-arbitrage formulas assume:

- no transaction costs or taxes;
- borrowing and lending at the same risk-free rate;
- ability to short sell when needed;
- known income, storage costs, and interest rates;
- no default risk.

These assumptions are deliberately strong. They identify the clean benchmark.

No Income

For a non-income-paying investment asset, the no-arbitrage forward price is

$$F_0 = S_0 e^{rT}.$$

Example 14 (Forward price with no income)
A non-dividend-paying stock has price $S_0 = 80$, the continuously compounded risk-free rate is $r = 4\%$, and maturity is $T = 0.75$ years. The forward price is

$$F_0 = 80e^{0.04\times 0.75} = 80e^{0.03} \approx 82.44.$$

Example 15 (Cash-and-carry arbitrage)
Suppose the same stock has a quoted forward price of 85. The no-arbitrage price is 82.44, so the forward is expensive. A trader can:

1. *borrow 80;*
2. *buy the stock;*
3. *short the forward at 85.*

At maturity, deliver the stock into the forward, receive 85, and repay

$$80e^{0.03} = 82.44.$$

The locked-in profit is

$$85 - 82.44 = 2.56.$$

Example 16 (Reverse cash-and-carry arbitrage)
If the quoted forward price is 81, the forward is cheap. A trader can short the stock, invest the proceeds, and enter a long forward. At maturity, the investment is worth 82.44. The trader buys the stock through the forward for 81 and returns it to close the short sale. The profit is

$$82.44 - 81 = 1.44.$$

Known Income

If the asset provides known income with present value I during the life of the forward, the forward price is

$$F_0 = (S_0 - I)e^{rT}.$$

The income lowers the forward price because the forward buyer does not receive income paid before delivery.

Example 17 (Forward price with known cash income)
A stock price is 100. It will pay a known dividend of 3 in six months. The one-year continuously compounded risk-free rate is 5%. The present value of the dividend is

$$I = 3e^{-0.05\times 0.5} \approx 2.93.$$

The one-year forward price is

$$F_0 = (100 - 2.93)e^{0.05} \approx 102.05.$$

Known Yield

If the asset provides a known continuous yield q, the forward price is

$$F_0 = S_0 e^{(r-q)T}.$$

Stock indices and foreign currencies are often modeled this way. For a stock index, q is the dividend yield. For a foreign currency, q is the foreign risk-free rate.

Example 18 (Index futures price)
An index is at 3,500. The risk-free rate is 4%, the dividend yield is 1.5%, and maturity is four months. With continuous compounding,

$$F_0 = 3{,}500e^{(0.04-0.015)(4/12)} = 3{,}500e^{0.008333} \approx 3{,}529.29.$$

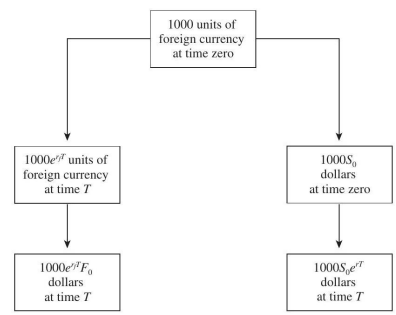

Figure 3: A foreign currency can be treated like an asset with a yield: foreign interest is the benefit from holding the currency.

Value of an Existing Forward Contract

The delivery price K is fixed in an existing forward contract. The current forward price F_0 is the delivery price that would make a new contract have zero value. For a long forward with maturity T,

$$f = (F_0 - K)e^{-rT}.$$

For a non-income-paying asset, this is equivalent to

$$f = S_0 - Ke^{-rT}.$$

With known income,

$$f = S_0 - I - Ke^{-rT}.$$

With known yield,

$$f = S_0 e^{-qT} - Ke^{-rT}.$$

Example 19 (Forward contract value after initiation)
A trader is long a one-year forward with delivery price $K = 50$. Six months later, the current six-month forward price for the asset is $F_0 = 54$ and the six-month risk-free rate is 4% continuously compounded. The value of the long forward is

$$f = (54 - 50)e^{-0.04\times 0.5} = 4e^{-0.02} \approx 3.92.$$

The contract is valuable because it allows the trader to buy at 50 when the current forward price is 54.

Futures Prices Versus Forward Prices

For many purposes, especially with deterministic interest rates, futures and forward prices are treated as equal. Daily settlement is the key difference. Futures gains and losses are realized every day, while forward gains and losses accumulate until maturity.

When interest rates are uncertain:

- if futures prices tend to rise when interest rates rise, long futures gains can be reinvested at high rates, making futures prices tend to exceed forward prices;
- if futures prices tend to fall when interest rates rise, long futures losses occur when financing is expensive, making futures prices tend to be below forward prices.

> **Practical convention.** For short-dated contracts and many classroom applications, we use the same pricing formula for forwards and futures. The distinction matters more when interest rates are volatile and strongly correlated with futures price changes.

Commodity Forward and Futures Prices

Storage Costs

Storage costs act like negative income. If the present value of storage costs is U, the forward price for an investment commodity is

$$F_0 = (S_0 + U)e^{rT}.$$

If storage costs are proportional to the spot price at continuous rate u, then

$$F_0 = S_0 e^{(r+u)T}.$$

Example 20 (Commodity with fixed storage costs)
Silver is at USD 25 per ounce. Storage costs have present value USD 0.40 per ounce over the life of a nine-month contract. The risk-free rate is 5%. The forward price is

$$F_0 = (25 + 0.40)e^{0.05\times 0.75} \approx 26.37.$$

Example 21 (Commodity with proportional storage costs)
A commodity spot price is 60. The risk-free rate is 4%, storage costs are 2% per year proportional to the spot price, and maturity is one year. Then

$$F_0 = 60e^{(0.04+0.02)} \approx 63.71.$$

Convenience Yield

For a consumption commodity, physical ownership can provide benefits that a futures contract does not provide:

- ability to keep a production process running;
- ability to profit from temporary shortages;
- avoiding disruption from supply-chain problems;
- meeting customer demand immediately.

These benefits are summarized by the *convenience yield*.

With proportional storage cost u and convenience yield y, the common cost-of-carry expression is

$$F_0 = S_0 e^{(r+u-y)T}.$$

Example 22 (Convenience yield)
Oil has spot price 80. The risk-free rate is 5%, storage costs are 1% per year of spot value, and the convenience yield is 4%. For a six-month contract,

$$F_0 = 80e^{(0.05+0.01-0.04)(0.5)} = 80e^{0.01} \approx 80.80.$$

Without convenience yield, the price would be

$$80e^{(0.05+0.01)(0.5)} \approx 82.44.$$

The convenience yield reduces the futures price because physical ownership has a benefit.

Investment Commodities Versus Consumption Commodities

For an investment commodity such as gold, arbitrage can often pin down the forward price from spot price, interest rates, and storage

costs. For a consumption commodity such as oil or copper, arbitrage usually gives an upper bound rather than an exact price:

$$F_0 \leq (S_0 + U)e^{rT}.$$

If the futures price were above this bound, traders could buy and store the commodity and short futures. If the futures price is below the bound, the reverse trade may be impossible because holders of the physical commodity may not be willing to sell inventory and give up convenience benefits.

> **Important distinction.** A low futures price for a consumption commodity does not automatically imply arbitrage. The missing asset is the convenience benefit from physical inventory.

Cost of Carry

The cost of carry is the cost of financing and storing the asset, minus income earned on the asset. For an investment asset,

$$F_0 = S_0 e^{cT},$$

where c is the cost of carry.

For a consumption commodity with convenience yield y,

$$F_0 = S_0 e^{(c-y)T}.$$

Contango and Backwardation

A futures curve is in *contango* when futures prices tend to rise with maturity. It is in *backwardation* when futures prices tend to fall with maturity.

Contango is often associated with financing and storage costs exceeding convenience yield. Backwardation is often associated with high convenience yield, scarcity, or strong current demand for the physical commodity.

Example 23 (Reading a futures curve)
Suppose spot oil is 82, the three-month futures price is 81, and the one-year futures price is 77. The curve is downward sloping. One interpretation is that near-term physical oil has a high convenience yield: inventory is valuable today because users want immediate access to the commodity.

Expected Future Spot Prices

Futures Price Is Not Usually an Unbiased Forecast

It is tempting to say that the futures price equals the expected future spot price. This is not generally correct. Futures prices are pricing objects, not pure forecasts.

Let k be the required expected return on the underlying asset. A simple relationship is

$$F_0 = \mathbb{E}(S_T)e^{(r-k)T}.$$

Therefore:

Systematic risk	Relation between k and r	Implication
No systematic risk	$k = r$	$F_0 = \mathbb{E}(S_T)$
Positive systematic risk	$k > r$	$F_0 < \mathbb{E}(S_T)$
Negative systematic risk	$k < r$	$F_0 > \mathbb{E}(S_T)$

Example 24 (Positive systematic risk)
A stock index has positive systematic risk, so investors require an expected return above the risk-free rate. The futures price can be below the expected future index level even though there is no arbitrage. The gap compensates investors for bearing market risk.

Normal Backwardation and Contango

Keynes and Hicks emphasized hedging pressure. If producers are natural short hedgers and speculators take the long side, speculators may require a positive expected return. In that case the futures price may be below the expected future spot price. This is often called normal backwardation.

If consumers are natural long hedgers and speculators take the short side, the reverse can occur: the futures price may exceed the expected future spot price.

> **Do not confuse terms.** Backwardation and contango describe the shape of the futures curve across maturities. Normal backwardation and normal contango describe the relation between a futures price and the expected future spot price.

Commodity Behavior from Chapter 35

Agricultural Commodities

Agricultural commodities often have:

- strong seasonality;
- weather-driven supply shocks;
- high volatility before harvest;
- storage limits and quality deterioration;
- jumps when information about crop size changes.

Metals

Precious metals such as gold and silver are closer to investment assets because they are widely held for investment. Industrial metals such as copper are closer to consumption assets because users value physical inventory for production.

Energy Products

Energy markets are central for hedging because both prices and quantities are uncertain.

- Crude oil can be stored, but storage capacity and location matter.
- Natural gas demand is seasonal and weather-sensitive.
- Electricity is difficult to store economically, so spot prices can spike sharply when demand or supply surprises occur.

Example 25 (Electricity and spikes)
If a heat wave increases air-conditioning demand while generation capacity is tight, electricity spot prices can jump. A futures or forward contract hedges average price over a delivery period, but it may not perfectly hedge operational exposure to hourly spikes unless the contract is designed for that load shape.

Mean Reversion and Seasonality

Commodity prices often display mean reversion: high prices encourage supply and discourage demand, while low prices discourage production and encourage demand. Seasonality arises when supply or demand follows the calendar.

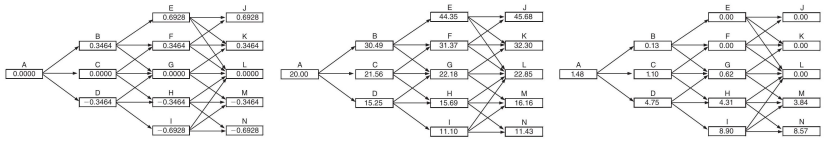

Figure 4: Commodity derivative models often use trees calibrated to futures prices. The economic inputs are not only volatility, but also term structure, seasonality, mean reversion, and convenience yield.

Modeling caution. Equity models often begin with geometric Brownian motion. Commodity models often need storage, seasonality, mean reversion, jumps, and convenience yields. The economic structure of the underlying asset matters.

Worked Problems

Example 26 (Short hedge effective price)
A firm shorts futures at 48.00 to hedge a sale. When the hedge is closed, the spot price is 46.80 and the futures price is 47.10. The basis is

$$b_2 = 46.80 - 47.10 = -0.30.$$

The effective sale price is

$$F_1 + b_2 = 48.00 - 0.30 = 47.70.$$

Example 27 (Long hedge effective price)
A firm goes long futures at 112.50. At the purchase date, the spot price is 118.00 and the futures price is 117.20. The final basis is

$$b_2 = 118.00 - 117.20 = 0.80.$$

The effective purchase price is

$$112.50 + 0.80 = 113.30.$$

Example 28 (Minimum-variance hedge ratio)
Spot price changes have standard deviation 2.4, futures price changes have standard deviation 3.0, and their correlation is 0.75:

$$h^* = 0.75\frac{2.4}{3.0} = 0.60.$$

If the exposure is 120,000 units and each futures contract is for 10,000 units,

$$N^* = 0.60\frac{120{,}000}{10{,}000} = 7.2.$$

The hedger would use about 7 contracts.

Example 29 (Equity hedge)
A portfolio is worth EUR 18 million and has beta 0.9. Index futures are quoted at 6,000 with multiplier EUR 10. One contract has value EUR 60,000. The number of contracts for a full market hedge is

$$N^* = 0.9\frac{18{,}000{,}000}{60{,}000} = 270.$$

The manager shorts 270 contracts.

Example 30 (Forward price with known yield)
An index is 1,200, the risk-free rate is 3.5%, the dividend yield is 2.0%, and maturity is nine months. Then

$$F_0 = 1{,}200e^{(0.035-0.020)(0.75)} = 1{,}200e^{0.01125} \approx 1{,}213.58.$$

Example 31 (Value of a forward on an income-paying asset)
A stock is 60. A dividend of 1.50 will be paid in three months. A long forward with six months remaining has delivery price 58. The continuously compounded risk-free rate is 4%. The present value of the dividend is

$$I = 1.50e^{-0.04(0.25)} \approx 1.485.$$

The value of the long forward is

$$f = S_0 - I - Ke^{-rT} = 60 - 1.485 - 58e^{-0.04(0.5)} \approx 1.66.$$

Example 32 (Storage cost and convenience yield)
A commodity spot price is 45. Financing cost is 4%, proportional storage cost is 3%, and convenience yield is 2%. For a one-year contract,

$$F_0 = 45e^{(0.04+0.03-0.02)} = 45e^{0.05} \approx 47.31.$$

Example 33 (Futures price and expected spot)
A commodity has expected future spot price $\mathbb{E}(S_T) = 100$, risk-free rate $r = 3\%$, required return $k = 7\%$, and maturity $T = 1$. Then

$$F_0 = 100e^{(0.03-0.07)} = 100e^{-0.04} \approx 96.08.$$

The futures price is below the expected future spot price because the asset has positive systematic risk.

Checklist for Students

After this lecture you should be able to:

- decide whether a short hedge or long hedge is appropriate;
- compute futures gains and losses for hedging examples;
- define the basis as $S_t - F_t$ and compute effective hedge prices;
- explain why basis risk remains after hedging;
- decompose basis risk in a cross hedge;
- derive and use $h^* = \rho\sigma_S/\sigma_F$;
- compute the optimal number of futures contracts;
- hedge an equity portfolio with stock-index futures;

- change a portfolio beta using futures;
- explain stack-and-roll hedging and roll risk;
- derive $F_0 = S_0 e^{rT}$ by no-arbitrage;
- price forwards with known income and known yield;
- value an existing forward contract;
- distinguish investment assets and consumption assets;
- use storage costs and convenience yield in commodity pricing;
- distinguish futures prices from expected future spot prices.

Practice Questions

1. A firm will sell 2 million gallons of a commodity in three months. The futures contract size is 50,000 gallons. Should the firm go long or short? How many contracts are needed for a one-to-one hedge?
2. A buyer will purchase 100,000 units in six months. It goes long futures at 18.40. At the purchase date the spot price is 20.10 and the futures price is 19.90. Compute the final basis and effective purchase price.
3. A seller shorts futures at 72. The final spot price is 70 and the final futures price is 69.50. Compute the effective sale price.
4. Explain why basis strengthening helps a short hedger but hurts a long hedger.
5. Spot-price changes have standard deviation 5, futures-price changes have standard deviation 4, and the correlation is 0.64. Compute the minimum-variance hedge ratio and hedge effectiveness.
6. A firm has exposure to 400,000 units. Each futures contract covers 20,000 units. If $h^* = 0.80$, compute the optimal number of contracts.
7. A portfolio is worth USD 75 million and has beta 1.10. Index futures are quoted at 4,800 with multiplier 250. How many contracts should be shorted to hedge market exposure?
8. The same portfolio manager wants to reduce beta from 1.10 to 0.60. How many contracts should be used?
9. A non-dividend-paying stock has $S_0 = 42$, $r = 5\%$, and $T = 0.5$. Compute the forward price.

10. If the quoted forward price in the previous question is 44.50, describe the arbitrage.

11. A stock is 90 and will pay a known dividend with present value 2. The risk-free rate is 4% and maturity is one year. Compute the forward price.

12. An index is 2,000, the risk-free rate is 3%, the dividend yield is 1.2%, and maturity is six months. Compute the futures price.

13. A long forward has delivery price 100. The current forward price for the remaining maturity is 108, the risk-free rate is 5%, and remaining maturity is one year. Compute the value of the long forward.

14. A commodity has spot price 35 and storage costs with present value 1.2. The risk-free rate is 6% and maturity is nine months. Compute the no-arbitrage forward price for an investment commodity.

15. A commodity has spot price 70, financing cost 4%, storage cost 2%, convenience yield 5%, and maturity one year. Compute the futures price using the cost-of-carry formula.

16. Explain why the reverse cash-and-carry arbitrage may fail for a consumption commodity.

17. Distinguish contango from normal contango.

18. If $\mathbb{E}(S_T) = 50$, $r = 3\%$, $k = 1\%$, and $T = 1$, compute F_0 and explain the sign of the risk premium.

19. Give two reasons electricity prices can be more volatile than crude oil prices.

20. Explain how weather can create both price risk and volume risk for an energy producer.